

Sprint 4: Executable Quote Engine

How a safe reference rate becomes an immutable customer offer

In progress

Starting status: Sprint 3 reference snapshots are delivered. Quote policy, calculation, storage, and API implementation have not started. The module specification and proposed economics decision are the first Sprint 4 deliverables.

Connection to Sprint 3

Sprint 3 stores an accepted reference rate and every provider fact behind it. Sprint 4 will select a fresh snapshot, add the configured spread and purchase fees, calculate exact fiat and crypto amounts, and freeze the result until expiry.

Planned Flow

```
accepted Sprint 3 reference snapshot
      + versioned quote policy
      + total fiat debit
        v
fees -> trade amount -> spread-adjusted rate
        v
immutable executable quote with short expiry
        v
Sendaza displays it; Sprint 6 later reserves inventory
```

Proposed Economic Rule

The request supplies total fiat debit. Fixed and percentage fees are deducted before conversion. Percentage fee rounds up to fiat atomic units; destination crypto rounds down to asset atomic units. The source snapshot expiry limits quote expiry. A quote never locks Sendaza money or reserves SLE inventory.

Planned Evidence

- Exact fixed, percentage, combined, and zero-fee tests.
- Limit, precision, rounding, and tiny-amount boundaries.
- Expiry and configuration-change evidence.

- Idempotency and concurrent creation tests.
- Real PostgreSQL atomicity and immutability tests.
- Private API contract and stable error tests.

Current Limitations

ADR-009 requires review. No quote table or endpoint exists yet. Production pricing remains inactive, so no executable quote can currently be issued.

Glossary

Total debit: all fiat the customer authorizes. **Trade amount:** total debit minus fees. **Spread:** configured adjustment from reference rate to customer rate. **Atomic unit:** smallest supported unit of a currency or asset. **Executable quote:** immutable short-lived economics that a later purchase may consume.

Started 1 September 2026. Sprint 4 remains in progress.